



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 3 • STANDARD 6.AT.3

Ratio Reasoning: Convert Measurements within the Same System

Lesson 3-6 • Teacher Copy

Name: Type your name

Date: Today's date

Class: Class period



TODAY'S OBJECTIVES

Content Objective: I can use ratio reasoning to convert between units within the same measurement system.

Language Objective: I can explain a conversion using the words unit ratio, convert, measurement system, and per.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Conversion factor

Unit ratio

Convert

Measurement system

Double number line

Ratio table

Customary units

Metric units



Tap any word to see its meaning and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

The ratio that trades one unit for another is a

2

A ratio comparing an amount to exactly unit is a unit ratio.

3

To change a measurement to another unit without changing the amount is to

4

Inches and feet belong to the customary

 ; centimeters and meters to the metric one.

5 Two lines whose matching ticks show the same amount in two units form a

 .

6 A table whose every column holds the same comparison is a

 .

7 Inches, feet, pounds, and gallons are .

8 Meters, grams, and liters are .

Write About the Math

How many cup-scoops of broth does the recipe need, and how do you know?

¿Cuántas tazas de caldo necesita la receta y cómo lo sabes?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Conversion factor**
Factor de conversión

☐ **Unit ratio**
Razón unitaria

☐ **Convert**
Convertir

☐ **Measurement system**
Sistema de medidas

☐ **Double number line**
Recta numérica doble

Model: Find the scale factor (120 divided by $8 = 15$), then multiply each ingredient by it. — Students find the scale factor of 15 and explain that multiplying both ingredients by 15 keeps the recipe proportional (5 cups tomatoes becomes 75 , 3 cups mozzarella becomes 45).

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The unit ratio is ____ cups to 1 quart. Cups are ____ than quarts, so I ____ to convert. $3 \times 4 =$ ____ cups. The amount of broth did not change — only the ____ did.
 - My answer is ____ because ____.
- Mi respuesta es ____ porque ____.*

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____.
 - My evidence is ____, so ____.
- Mi afirmación es ____.*
Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Conversion factor
2. **Notes 2:** Unit ratio
3. **Notes 3:** Convert
4. **Notes 4:** Measurement system
5. **Notes 5:** Double number line
6. **Notes 6:** Ratio table
7. **Notes 7:** Customary units
8. **Notes 8:** Metric units
9. **Watch for this mistake:** *A common mistake when converting within a measurement system is choosing the operation by habit instead of by unit size — multiplying every time. Converting 36 inches to feet by multiplying gives $36 \times 12 = 432$ feet, taller than a 40-story building. Ask first which unit is smaller: it takes MORE small units to make the same amount, so going to a smaller unit multiplies, and going to a larger unit divides.*
10. **Try It 1:** — *Every unit ratio says how many of the smaller unit fit inside exactly 1 of the larger one.*
11. **Try It 2:** — *Conversions in this lesson stay inside one system — customary with customary, metric with metric.*